The Clebsch orientation cubic: arithmetic covers and icosahedral harmonics

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Abstract

The Clebsch matching geometry has two orientations, first separated by a cubic observable. We determine the rational square class of Hitchin's ordered-icosahedron double cover: its function field is obtained by adjoining $\sqrt{5J_0}$, and on the Clebsch four-space $J_0 = 16\sigma_3^2$. The nonbranch Clebsch fibre is therefore the constant golden torsor $\operatorname{Spec}\mathbf{Q}(\sqrt{5})$. Its explicit exchanger has good reduction at 11, where its nonsquare spinor class is the finite Clebsch sheet involution. An equivariant argument places the signed matching cubic on the line $\langle\sigma_3\rangle$ over $\mathbf{F}_{11}$. In characteristic zero, the same integral cubic line is realized by the degree-six Gaunt invariant on the four-dimensional summand of the ten icosahedral face-axis harmonics, with exact restriction

$$\frac{1}{4\pi}\int_{S^2} F_y^3 = -\frac{784000}{1247103}\sigma_3(y).$$

Thus one cubic line links the arithmetic orientation fibre, its finite matching specialization, and a standard spherical-harmonic observable.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; spherical harmonics; bond-orientational order.

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1 Introduction

The Clebsch matching configuration has two sheets. Forgetting the sheet retains the unordered matching data; recovering it requires an odd observable. Over $\mathbf{F}_{11}$ that first observable is a signed cubic tensor on a ten-dimensional matching quotient. We ask whether the orientation and its cubic have a characteristic-zero source.

Hitchin's incidence between harmonic cubics and ordered icosahedra supplies the double cover. Let H be the rational seven-space of harmonic cubics, let J_0 denote Hitchin's invariant sextic, and write

$$V = \{(y_1, \dots, y_5) : \textstyle\sum_i y_i = 0\}, \qquad \sigma_3(y) = \tfrac{1}{3}\textstyle\sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic. The following theorem gives the arithmetic and harmonic realizations of the same cubic line.

Theorem 1.1 (Arithmetic orientation and harmonic realization)**.** `MAIN-1`.

1. *The function field of Hitchin's ordered-icosahedron incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}), \qquad J_0|_V = 16\sigma_3^2.$$

Hence its restriction to $D(\sigma_3) \subset \mathbf{P}(V)$ is the constant golden torsor

$$D(\sigma_3) \times_{\operatorname{Spec}\mathbf{Q}} \operatorname{Spec}\mathbf{Q}(\sqrt{5}) \longrightarrow D(\sigma_3).$$

The explicit fibre over xyz has good reduction at 11: its Galois exchanger reduces to the nontrivial class

$$T_{11} = \mathrm{PGL}_2(11)/\,\mathrm{PSL}_2(11),$$

while an explicit A_5-equivariant identification places the signed matching cubic on the line $\langle\sigma_3\rangle$.

2. *The zonal degree-six harmonics on the ten axes through opposite icosahedral faces contain V as their Petersen (-2)-eigenspace. Label the unit face axes by pairs $\{i,j\}$, and put*

$$F_y(\omega) = \sum_{i<j}(y_i + y_j)P_6(u_{ij}\cdot\omega), \qquad P_6(s) = \frac{231s^6 - 315s^4 + 105s^2 - 5}{16}.$$

On this four-space the spherical Gaunt cubic is

$$\frac{1}{4\pi}\int_{S^2} F_y(\omega)^3\, d\omega = -\frac{784000}{1247103}\sigma_3(y).$$

Equivalently, the Clebsch cubic line is the four-channel restriction of the standard degree-six Steinhardt cubic.

The first assertion determines the rational square class of Hitchin's known degree-two cover by evaluating its golden fibre. It does not claim an integral incidence model at every prime: the abstract equation $z^2 = 5J_{\mathbf{Z}}$ is etale on $D(10J_{\mathbf{Z}})$, while comparison with the geometric incidence variety is proved only after inverting an unspecified integer. The statement at 11 concerns the explicitly integral golden fibre and exchanger, so it does not depend on a global good-reduction claim.

The second assertion gives an exact analytic realization, not an empirical materials claim. The ordinary uniformly weighted icosahedron lies in the trivial summand; the orientation cubic lives on a nonuniform four-dimensional decoration of the face axes. Moreover, the displayed characteristic-zero scalar has denominator divisible by 11. The cross-characteristic conclusion is equality of the integral line $\langle\sigma_3\rangle$, not reduction of one scalar.

The proof has four steps. Section 2 fixes the square class of the rational cover and separates its three integral boundaries. Section 3 constructs the equivariant finite tensor bridge. Section 4 computes the golden exchanger and its spinor class at 11. Section 5 constructs the degree-six channel and evaluates its Gaunt cubic. Section 7 records which steps are conceptual or literature-backed and which explicit data receive independent exact audits.

2 Orientation as a quadratic cover

Let R be a domain with an involution ι. Write

$$R^+ = \{r : \iota(r) = r\}, \qquad R^- = \{r : \iota(r) = -r\}.$$

The elementary algebra behind the construction is that one nonzero odd element trivializes the odd summand after localization.

Lemma 2.1 (Localized odd generator)**.** `ALG-1`*. Assume that* 2 *is invertible in* R*, and let* $C \in R^-$ *be nonzero. After localizing at* C^2*,*

$$R[(C^2)^{-1}] = R^+[(C^2)^{-1}] \oplus C\,R^+[(C^2)^{-1}].$$

Proof. Every $r \in R$ has the decomposition

$$r = \frac{r + \iota(r)}{2} + \frac{r - \iota(r)}{2}.$$

The first term is invariant. If r is odd, then r/C is invariant after C is inverted, because both numerator and denominator change sign. Since inverting C^2 also makes C invertible, the odd summand is C times the invariant summand. The intersection is zero because an element in it is both invariant and odd. □

Thus an orientation-forgetting quotient is generically quadratic whenever its odd algebra is generated by one element. For the sign-twisted icosahedral action on the Clebsch four-space, the first candidate odd generator is the cubic

$$\sigma_3(y) = \frac{1}{3}\sum_{i=1}^{5} y_i^3, \qquad \sum_{i=1}^{5} y_i = 0.$$

Here is the precise characteristic boundary. Put

$$A_R = R[y_1, \dots, y_5]/(e_1), \qquad \Delta = \prod_{i<j}(y_i - y_j),$$

where $R = \mathbf{Z}[1/30]$. The alternating group permutes the coordinates, and an odd permutation τ defines the twisted involution $\kappa(y_i) = -y_{\tau(i)}$. Nonmodular invariant theory gives

$$A_R^{A_5} = R[e_2, e_3, e_4, e_5, \Delta]/(\Delta^2 - \operatorname{disc}(e_2, e_3, e_4, e_5)).$$

Indeed, an A_5-invariant polynomial is the sum of a symmetric polynomial and Δ times a symmetric polynomial; exactness of invariants after imposing $e_1 = 0$ uses $60 \in R^\times$. Moreover

$$\kappa(e_i) = (-1)^i e_i, \qquad \kappa(\Delta) = -\Delta.$$

Thus $e_3 = \sigma_3$ is the first nonconstant odd invariant. On $D(e_3)$, Lemma 2.1 identifies the full invariant algebra with its even part plus e_3 times its even part. This statement does not assert the same presentation in characteristics $2, 3, 5$; modular invariants can appear there. Hitchin's binary-sextic realization has the same nonmodular boundary: the classical generators and the sextic J_6 used below are available uniformly in characteristic greater than 5 [2, 5]. Dye's earlier finite-geometry theorem already gives the complementary field-level criterion: Clebsch hexagons exist precisely when the characteristic is not 2 and 5 is a square [3, Theorem 1].

The threefold in the next statement originates with Mukai and Umemura; Hitchin's Grassmannian model supplies the incidence calculation used here [4, 2].

Theorem 2.2 (Rational Hitchin–Clebsch cover)**.** `HC-1`*. Let* H *be the rational seven-space of harmonic cubics, and let* $\mathcal{I}_{\mathbf{Q}}$ *be the incidence variety of pairs* $([f], U)$*, where* U *is a Mukai–Umemura isotropic three-plane and* $f \perp U$*. The induced extension of* $\mathbf{Q}(\mathbf{P}(H))$ *is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

On the Clebsch four-space,

$$J_0 = 16\sigma_3^2,$$

so over $D(\sigma_3)$ *the cover is the constant golden torsor* $\operatorname{Spec}\mathbf{Q}(\sqrt{5})$*.*

Proof. Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension. Hitchin identifies its branch hypersurface with the irreducible sextic $J_0 = 0$, and computes $J_0|_{\mathbf{P}(V)} = 16\sigma_3^2$ [1, §§9–10]. Since $\mathrm{Pic}(\mathbf{P}(H)) = \mathbf{Z}$ has no two-torsion, the extension therefore has the form $\mathbf{Q}(\mathbf{P}(H))(\sqrt{cJ_0})$ for one rational square class c.

For $f = xyz$, Hitchin exhibits the two icosahedra with golden coordinates $t^2 - t - 1 = 0$. The value $J_0(f)$ is a nonzero rational square by the displayed Clebsch restriction. Hence this fibre determines $c = 5$. Dividing by $4\sigma_3$ gives the final assertion. $\square$

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \qquad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 2.2. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N, but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

3 The Clebsch cubic bridge

Let G be the order-60 stabilizer of the H_3-invariant matching in $\mathrm{PGL}_2(\mathbf{F}_{11})$, and let Ω be its 22-point matching orbit. For a reference matching M_0, put

$$W = \langle q_M - q_{M_0} : M \in \Omega \rangle .$$

The two $\mathrm{PSL}_2(\mathbf{F}_{11})$-sheets define signs $\epsilon(M) \in \{\pm 1\}$ and the symmetric tensor

$$T = \sum_{M \in \Omega} \epsilon(M) [q_M - q_{M_0}]^{\otimes 3} \in \mathrm{Sym}^3(W).$$

Theorem 3.1 (Finite tensor bridge)**.** `HC-3`. *There is a G-equivariant isomorphism*

$$S : \mathbf{F}_{11}^{\binom{[5]}{2}} \longrightarrow W$$

such that, after using the standard pairing to identify the permutation module with its dual and restricting along

$$E(y)_{\{i,j\}} = y_i + y_j, \qquad \sum_i y_i = 0,$$

one has

$$E^*(S^{-1})^{\otimes 3} T = 4\sigma_3.$$

In particular, the signed matching cubic and the Clebsch cubic span the same nonzero invariant line over $\mathbf{F}_{11}$.

Proof. Conjugation on the normalizers of the five Klein four subgroups identifies G with A_5 in its degree-five action. Hence it acts on the ten two-subsets. The matching construction gives a second ten-dimensional representation on W. On the classes $1, 2A, 3A, 5A, 5B$, both characters are

$$(10, 2, 1, 0, 0),$$

so both modules decompose as $\mathbf{1} \oplus V_4 \oplus V_5$. This proves that an equivariant isomorphism exists. To fix its three irreducible scalings, we solve $\rho_W(g)S = S\rho_P(g)$. The recorded invertible solution satisfies this identity for all 60 elements.

The transported restriction of T is an A_5-invariant cubic on V_4. Such cubics form one line: the degree-three symmetric orbit sums on five variables have exponent types 3, $2+1$, and $1+1+1$, and after imposing $\sum y_i = 0$ only the line generated by σ_3 remains. It therefore suffices to determine one scalar. One mixed polarized entry of the transported tensor is 6, while the corresponding entry of σ_3 is $-1/3 = 7$. Since $4 \cdot 7 = 6$ in $\mathbf{F}_{11}$, the scalar is 4.

The displayed character argument and one-entry calculation are the proof mechanism. A primary exact calculation and an independent reconstruction audit the chosen intertwiner on all 60 elements and all 64 contracted entries. Lean optionally checks the final literal tensor equality; no formal result is used for the representation-theoretic reduction. □

Remark 3.2 (Normalization). The characteristic-zero Gaunt scalar in Theorem 5.1 has denominator divisible by 11. The theorem therefore identifies a common integral cubic line; it is not obtained by reducing that rational scalar modulo 11.

4 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz. Its two ordered icosahedra are Galois conjugate over $\mathbf{Q}(\sqrt{5})$; modulo 11 they become two rational Clebsch parents, and the Galois exchange becomes the nontrivial spinor class. We prove these statements before introducing the Mathieu carrier.

4.1 The golden fibre

Let $t^2 - t - 1 = 0$, and put

$$I_t = \{[0:t:1], [0:t:-1], [1:0:t], [-1:0:t], \\ [t:-1:0], [-t:-1:0]\} \subset \mathbf{P}^2.$$

These six points represent the six opposite vertex pairs of an icosahedron.

Proposition 4.1 (Golden fibre). `HC-2`. *Over a field of characteristic different from* 2 *and* 5*, the points of* I_t *lie on* $xyz = 0$*, have a common nonzero norm, have normalized squared inner product* $1/5$*, and form a six-arc. The two configurations over* $\mathbf{Q}$ *are therefore defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's normalized cover of ordered icosahedra they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t+1$, its squared norm is $t^2+1 = t+2$. The inner product of two distinct representatives is $\pm t$, while

$$(t+2)^2 = 5(t+1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point determinants are $\pm 2t$ or $\pm 2t^2$, so none vanishes under the stated hypotheses. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. $\square$

4.2 The exchanger and the tetrahedral hinge

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \qquad R^2 = \mathrm{diag}(1, -1, -1), \qquad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two ordered icosahedra.

Proposition 4.2 (Tetrahedral hinge)**.** `HC-6`. *Let* $G_t = \mathrm{Stab}_{\mathrm{SO}_3}(I_t) \cong A_5$*. Then*

$$RG_tR^{-1} = G_{1-t}, \qquad G_t \cap G_{1-t} \cong A_4.$$

Proof. The first equality follows from $R(I_t) = I_{1-t}$. A direct linear calculation shows that the cubics through $I_t \cup I_{1-t}$ form the one-dimensional space spanned by xyz: write a general ternary cubic as

$$ax^3 + bx^2y + cx^2z + dxy^2 + exyz + fxz^2 + gy^3 + hy^2z + iyz^2 + jz^3.$$

Substitution of the three opposite pairs, followed by $t^2 = t+1$, gives

$$g = i = 0, \quad h(t+1) + j = 0, \quad c + j(t+1) = 0,$$

$$a + f(t+1) = 0, \quad b(t+1) + g = 0, \quad a(t+1) + d = 0.$$

Equating the rational and t-parts kills every coefficient except e. Thus every element of the intersection fixes the plane-triple cubic.

The rotations given by even signed coordinate permutations fix xyz and each I_t; they form the tetrahedral group A_4. Conversely, the icosahedral group $G_t \cong A_5$ acts on its five plane-triple cubics. This is its standard transitive action of degree five, whose point stabilizer is A_4. Hence the common stabilizer is exactly the displayed tetrahedral subgroup. $\square$

4.3 Reduction modulo eleven

The roots of $t^2 - t - 1$ in $\mathbf{F}_{11}$ are 4 and 8. For the working Clebsch arc

$$\begin{aligned} A = \{&[1:10:0], [1:9:1], [1:4:7], \\ &[1:8:5], [0:1:4], [1:1:7]\}, \end{aligned}$$

define

$$M_4 = \begin{pmatrix} 1 & 4 & 7 \\ 5 & 10 & 3 \\ 1 & 8 & 1 \end{pmatrix}, \qquad M_8 = \begin{pmatrix} 4 & 5 & 5 \\ 9 & 10 & 7 \\ 4 & 7 & 10 \end{pmatrix}.$$

Six projective substitutions give

$$M_4 I_4 = A, \qquad M_8 I_8 = A,$$

and $\det M_4 = 1,\ \det M_8 = 9$. Furthermore,

$$M_8^{-1} M_4 = 3R$$

in $\mathrm{GL}_3(\mathbf{F}_{11})$. Thus forgetting the ordered parent identifies both configurations with the same unmarked Clebsch arc, while their relative projectivity is the reduction of the golden exchanger.

Proposition 4.3 (Spinor specialization)**.** `HC-4`. *The reduction of R represents the nontrivial element of*

$$T_{11} = \mathrm{PGL}_2(11)/\,\mathrm{PSL}_2(11).$$

Proof. For $Q(x,y,z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz-plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } \mathbf{F}_{11}^{\times}/\mathbf{F}_{11}^{\times 2}.$$

The class of 2 is nonsquare modulo 11. The standard identifications

$$\mathrm{SO}_3(11) \cong \mathrm{PGL}_2(11), \qquad \Omega_3(11) \cong \mathrm{PSL}_2(11)$$

identify the spinor quotient with T_{11}, proving the claim. □

4.4 The marked Mathieu carrier

The ternary Golay–Hadamard model supplies two explicit subgroups

$$H_{\mathrm{row}}, H_{\mathrm{col}} \cong M_{11} \quad \text{inside} \quad M_{12}.$$

The order-twelve Hadamard model and its row–column outer automorphism have a direct construction in [6, §4]; the two M_{11} maximal-subgroup classes and the $L_2(11)$ class are recorded in the online ATLAS [7]. One is the pure coordinate group of the signed code and the other is the stabilizer of the parity coordinate. Their exact twelve-point generators give

$$H_{\mathrm{row}} \cap H_{\mathrm{col}} = \mathrm{PSL}_2(11).$$

Consequently

$$|H_{\mathrm{row}} H_{\mathrm{col}}| = \frac{7920^2}{660} = 95040 = |M_{12}|,$$

so the two parents generate M_{12}. Hadamard row–column duality induces the noninner automorphism of M_{12} and exchanges their two conjugacy classes. The finite assertions in this paragraph are checked from the displayed permutation carriers by the primary certificate and an implementation-independent replay described in Section 7.

There is no canonical unmarked identification between the golden and Mathieu pairs. The precise conclusion is the following marked statement.

Corollary 4.4 (Marked Hitchin–Mathieu torsor)**.** *Choose the marking $I_4 \leftrightarrow H_{\mathrm{row}}$. There is a unique equivariant bijection*

$$\{I_4, I_8\} \longrightarrow \{H_{\mathrm{row}}, H_{\mathrm{col}}\}$$

that extends this choice. It sends I_8 to H_{col}, and intertwines the reduced Hitchin exchanger with Hadamard row–column duality. Without the marking there are exactly two such equivariant bijections and neither parent is distinguished.

Proof. Both sets are free transitive C_2-sets: Proposition 4.3 supplies the nontrivial exchange on the Hitchin side, and Hadamard duality supplies it on the Mathieu side. The image of one marked element determines an equivariant map between free transitive C_2-sets. There are two possible markings. □

This marked, and only marked, compatibility is `HC-5`.

The certificate for this section checks only the explicit substitutions, matrix identities, permutation-group orders, intersection, and carrier exchange. Propositions 4.1–4.3 and Corollary 4.4 are the mathematical proof; the certificate is not used to infer the golden field, the A_4 hinge, the spinor formula, or the marked-torsor lemma.

Integral boundary. Over a finite field of characteristic different from 2 and 5, the abstract quadratic cover gives

$$\#X_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R; we do not replace the finite exceptional set by $p \neq 2, 5$.

5 The degree-six harmonic realization

Let $u_e \in S^2$, $1 \leq e \leq 10$, be the unoriented axes through pairs of opposite icosahedral faces. The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \qquad L(a)(\omega) = \sum_e a_e P_6(u_e \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16}\left(231s^6 - 315s^4 + 105s^2 - 5\right).$$

The degree-six bond-orientational invariant is the first standard icosahedral channel in the Steinhardt–Nelson–Ronchetti construction [8, 9]. The following theorem identifies its four-dimensional symmetry-breaking component.

Theorem 5.1 (Harmonic Clebsch bridge)**.** `PH-1`, `PH-2`, *and* `PH-3`. *Let A be the Petersen adjacency matrix on the ten face axes.*

1. *The map L is injective, and its Gram matrix is*

$$K = \frac{1}{243}(196I + 47J - 112A).$$

Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{81}, \qquad \frac{140}{81}, \qquad \frac{28}{81}.$$

2. *Label the axes by the pairs* $\{i,j\} \subset \{1,\dots,5\}$. *The Clebsch summand is the Petersen* (-2)*-eigenspace*

$$a_{ij} = y_i + y_j, \qquad \sum_i y_i = 0.$$

3. *For*

$$F_y(\omega) = \sum_{i<j} (y_i + y_j) P_6(u_{ij} \cdot \omega),$$

the spherical cubic is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 \, d\omega = -\frac{784000}{1247103} \, \sigma_3(y).$$

If $F = \sum_{m=-6}^{6} q_{6m} Y_{6m}$ *in the orthonormal Condon–Shortley convention and*

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1} q_{6m_2} q_{6m_3},$$

then

$$\int_{S^2} F^3 \, d\omega = -\frac{130}{\sqrt{3553\pi}} \, W_6(F).$$

Proof. The face-axis permutation representation decomposes as $\mathbf{1} \oplus V_4 \oplus V_5$, while

$$\mathcal{H}_6|_{A_5} \simeq \mathbf{1} \oplus V_3 \oplus V_4 \oplus V_5.$$

Direct evaluation of $P_6(u_e \cdot u_f)$ gives the displayed Gram matrix. The Petersen eigenvalues $3, 1, -2$, with multiplicities $1, 5, 4$, give the three positive Gram eigenvalues and prove injectivity. For $a_{ij} = y_i + y_j$,

$$(Aa)_{ij} = \sum_{\{k,l\} \cap \{i,j\} = \varnothing} (y_k + y_l) = -2(y_i + y_j),$$

so these vectors form the four-dimensional eigenspace.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi} \int F_y^2 = \frac{2800}{351}, \qquad \frac{1}{4\pi} \int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the third displayed scalar. These moments follow by expanding P_6 and applying

$$\frac{1}{4\pi} \int_{S^2} \omega_1^{2a} \omega_2^{2b} \omega_3^{2c} \, d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

any monomial with an odd exponent integrates to zero. Thus the scalar is recoverable directly from the displayed axes and this moment formula.

Finally, the Gaunt formula expresses the triple integral in the orthonormal spherical-harmonic basis as a product of two Wigner $3j$-symbols [10, §§14.30 and 34.3(vii)]. Here

$$\begin{pmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{pmatrix} = -\frac{20}{\sqrt{46189}},$$

which gives the stated conversion to W_6. The exact axis, Gram, moment, and Wigner calculations are checked by two independent implementations. □

Order-parameter meaning. A uniform icosahedral neighbour cage lies in the trivial summand. The Clebsch cubic instead sees the V_4 component of nonuniform weights on the ten opposite face pairs. For a face-axis weight vector a, the projector

$$P_{V_4} = \frac{(A-3I)(A-I)}{15}$$

extracts this component. A scale-free signed descriptor is then

$$C_4(a) = \frac{\langle L(P_{V_4}a)^3\rangle}{\langle L(P_{V_4}a)^2\rangle^{3/2}}, \qquad \langle f\rangle = \frac{1}{4\pi}\int_{S^2} f.$$

For the witness in the proof,

$$C_4 = -\frac{120\sqrt{273}}{3553}.$$

Possible weights include opposed-face distortion, chemical decoration, or normal stress. Whether C_4 predicts rearrangement, yielding, or nucleation beyond standard descriptors is an empirical question, recorded as `PH-4`. No applied performance claim is made here.

Remark 5.2 (Which symmetry flips the sign). The cubic is invariant under the rotational A_5-action. The sheet exchange is the odd element in the sign-twisted extension of this action, not an ordinary rotation of a uniformly weighted icosahedron. Any physical realization must therefore specify the decorated face-axis state and the outer-odd operation; the existence of an icosahedral cage alone does not produce the bit.

6 Consequences and boundaries

The two legs of Theorem 1.1 meet on the integral line $\langle\sigma_3\rangle$. This has three immediate consequences.

1. The finite matching tensor and the degree-six Gaunt cubic are two realizations of one A_5-invariant cubic line, although their displayed scalars cannot be reduced into one another at 11.
2. The two golden icosahedral parents form the fibre of one arithmetic torsor. Their common stabilizer is the tetrahedral group A_4, and their exchanger becomes the finite Clebsch sheet bit.
3. After choosing one parent, Corollary 4.4 identifies the golden pair with the two Mathieu M_{11} carriers. There is no canonical identification without this marking.

Theta, Fourier, quantum, four-sheet holonomy, and degree-23 coherence comparisons are not part of the present theorem complex. They enter only if they can be derived from the orientation cover with precise hypotheses. Otherwise they remain research inventory.

7 Verification and trust

The proof surface distinguishes conceptual arguments, classical inputs, and exact audits. Human proofs carry the theorem; certificates check only the displayed finite matrices, carrier data, and arithmetic constants. The machine-readable ledger in `verification/trust_manifest.json` records the proof mode and evidence for every manuscript claim.

For the finite tensor bridge, invariant theory reduces the conclusion to one nonzero scalar on the unique cubic line. The displayed intertwiner and one witness determine that scalar;

two exact implementations audit the 22 orbit vectors, the sheet signs, all 60 equivariance equations, and the contraction. A Lean terminal separately checks the final literal 4^3-tensor equality over $\mathbf{F}_{11}$, but it is not part of the release gate and does not formalize the geometric or representation-theoretic argument.

The harmonic bridge likewise has a primary exact generator, a canonical JSON certificate, and a separate replay. Both reconstruct the ten face axes over $\mathbf{Q}(\sqrt{5})$, the full Gram and Petersen matrices, and the quadratic and cubic spherical moments. The replay separately checks the Petersen minimal polynomial and the factorial formula for the Wigner $3j$-symbol. Neither route uses floating-point arithmetic.

The arithmetic specialization is proved in the text. Its compact certificate checks only the six projective substitutions, the comparison and reflection matrices, and the explicit twelve-point Mathieu carriers. The independent replay reimplements those checks and enumerates the two M_{11} parents, their order-660 intersection, their M_{12} join, and the carrier exchange without importing the primary program. Neither route is used to infer the golden field, the A_4 hinge, the spinor formula, or the marked-torsor lemma. The exact commands are

```
python3 verification/evidence/arithmetic_cover.py --check
python3 verification/evidence/arithmetic_cover_replay.py
sha256sum -c verification/evidence/arithmetic_cover.sha256
```

run from the paper directory. The integral model and novelty claims are established by the source audit and the scheme-theoretic argument in Section 2, not by a certificate. That argument proves the rational 5-twist and the abstract integral quadratic algebra, while retaining an unspecified finite bad set for comparison with the geometric incidence model.

The aggregate runner checks the frozen identity of all nine theorem-like statements, the claim-ledger correspondence, the primary and independent exact audits, and a warning-free manuscript build. The command

```
python3 verification/verify_release.py
```

is run from the paper directory. The empirical descriptor row `PH-4` remains inventory and is not a premise of Theorem 1.1.

8 Conclusion

The Clebsch orientation is governed by one cubic line in three settings. Over $\mathbf{Q}$, it trivializes Hitchin's cover to the golden torsor on the nonbranch Clebsch chart. Over $\mathbf{F}_{11}$, the explicit golden exchanger has the nonsquare spinor class and the signed matching tensor is $4\sigma_3$. Over $\mathbb{R}$, the same line is the four-channel restriction of the degree-six Gaunt cubic.

The integral boundary matters. The abstract quadratic algebra is explicit, but the geometric incidence comparison is not known at every prime outside characteristics 2 and 5, and the characteristic-zero Gaunt scalar does not reduce modulo 11. What survives both boundaries is the line $\langle\sigma_3\rangle$: the first odd invariant that records which orientation was forgotten.

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